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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
FALL SEMESTER 2025-2026

SLOT: G2

Programme Name & Branch : B.Tech- CSE and Specialization Programmes
Course Code and Course Name : BCSE355L – Cloud Architecture Design
Date of Examination : 11-OCT-2025
Exam Duration : 90 minutes **Maximum Marks: 50**

General instruction(s): Answer All Questions

CO2- Gain expertise in designing and implementing highly available, scalable, and Fault-tolerant AWS solutions.
CO3- Learn best practices for security, compliance, and cost optimization in AWS Architecture.
CO4- Develop proficiency in selecting appropriate AWS services and resources to meet specific business requirements

Q. No	Question	M	CO	BL
1.	A university hosts its student portal (vtop.vit.ac.in) and internal admin tools on AWS within a VPC containing two subnets: a public subnet (10.0.1.0/24) for the student portal and an admin subnet (10.0.2.0/24) for internal tools. To secure access, the network administrator must configure Network Access Control Lists (NACLs) such that the public subnet allows inbound HTTP traffic on port 80 from IP range 203.0.113.0/24 and SSH traffic on port 22 from 198.51.100.0/24, while the admin subnet allows only HTTPS traffic on port 443 from the campus admin network 192.0.2.0/24. All other inbound traffic to both subnets must be explicitly denied. How should the NACLs be configured to meet these requirements? Illustrate your answer with rule tables and diagram.	10	CO4	3
2.	As the cloud infrastructure lead for a digital banking platform experiencing outages and performance issues, how would you redesign the Amazon RDS architecture to ensure high availability during infrastructure failures, enable fast recovery from data loss, optimize read/write performance under load, and maintain secure the sensitive financial data? Suggest and justify a revised RDS architecture that meets the above given requirements with suitable illustrated diagrams.	10	CO4	2
3.	Given the PlayerStats table with PlayerID as the primary key and attributes like GameTitle, Timestamp, and HighScore, how would you use DynamoDB's indexing to meet two distinct query needs: (i).To find all achievements for a specific player sorted by Timestamp (ii).To create a global leaderboard by querying for a specific GameTitle and sorting by HighScore? Explain which type of index, a Local Secondary Index or a Global Secondary Index, is appropriate for each scenario and why, including a summary of the fundamental differences between the two. Use diagrams with table schema for highlighting the concept of LSI and GSI.	10	CO4	3
4.	Your task is to use AWS Cloud Watch tool that can address the company's monitoring and performance management needs for their new video-on-demand service. The service is experiencing intermittent issues, and there's a lack of centralized monitoring for its EC2 instances and S3 buckets. Your justification must explain how the recommended tool would collect metrics, set up alarms for automatic notifications, provide a unified typical diagram for	10	CO3	3

	visualization, and assist in log analysis to pinpoint the root cause of application crashes.			
5.	A global retail company needs to manage and automate routine administrative tasks on thousands of its EC2 instances and on-premises servers. The company faces challenges with consistent security patching, configuration management, secure access without SSH, and the automation of repetitive operational tasks. Identify and describe five specific features of AWS Systems Manager that can be used to address these problems.	10	CO3	4

ANSWER KEY

1.NACL

This setup ensures that each subnet is protected according to its role, with precise control over inbound traffic and compliance with best practices for subnet-level security using NACLs

1. Public Subnet (10.0.1.0/24) – Student Portal

Inbound Rules:

Rule #	Type	Protocol	Port Range	Source IP	Allow/Deny
100	HTTP	TCP	80	203.0.113.0/24	ALLOW
110	SSH	TCP	22	198.51.100.0/24	ALLOW
*	ALL	ALL	ALL	0.0.0.0/0	DENY

Outbound Rules:

Rule #	Type	Protocol	Port Range	Destination IP	Allow/Deny
100	ALL	ALL	ALL	0.0.0.0/0	ALLOW

- Outbound rule allows responses to HTTP and SSH requests.
- NACLs are stateless, so return traffic must be explicitly allowed.

🔷 2. Admin Subnet (10.0.2.0/24) – Internal Tools

Inbound Rules:

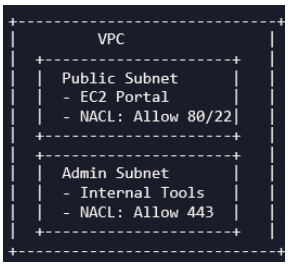
Rule #	Type	Protocol	Port Range	Source IP	Allow/Deny
100	HTTPS	TCP	443	192.0.2.0/24	ALLOW
*	ALL	ALL	ALL	0.0.0.0/0	DENY

Outbound Rules:

Rule #	Type	Protocol	Port Range	Destination IP	Allow/Deny
100	ALL	ALL	ALL	0.0.0.0/0	ALLOW

> 🔸 This configuration ensures only secure HTTPS access from the campus admin network.
> 🔸 All other traffic is denied by the catch-all rule.

Conceptual Diagram



2. RDS

Architecture Components

1. Multi-AZ Deployment (High Availability)

- Deploy Amazon RDS in Multi-AZ mode.

- AWS automatically provisions a standby replica in a separate Availability Zone.

- In case of failure, automatic failover ensures minimal downtime.

Ensures resilience against AZ-level outages

2. Read Replicas (Performance Optimization)

- Create Read Replicas in the same or different regions.

- Route read-heavy queries (e.g., transaction history, reports) to replicas.
- Use Amazon Aurora for faster replication and auto-scaling if needed.

Offloads read traffic and improves throughput

3. Automated Backups & Manual Snapshots (Recovery)

- Enable automated backups with a retention policy (e.g., 7–35 days).
- Schedule manual snapshots before major updates.
- Use Point-in-Time Recovery to restore to any moment within the backup window.

Enables fast recovery from logical errors or data corruption

4.Encryption & Access Control (Security)

- Enable encryption at rest using AWS KMS.
- Use SSL/TLS for encryption in transit.
- Restrict access using IAM roles, security groups, and parameter groups.
- Enable database-level auditing for compliance.

Protects sensitive financial data and meets regulatory standards

Feature	AWS Service / Strategy	Benefit
High Availability	RDS Multi-AZ	Automatic failover, minimal downtime
Performance Optimization	Read Replicas / Aurora	Load balancing, faster reads
Data Recovery	Automated Backups / Snapshots	Point-in-time restore
Security	KMS, SSL, IAM, Security Groups	Encryption, access control

3. LSI /GSI

Base Table: PlayerStats

Attribute	Type	Description
PlayerID	Partition Key	Unique identifier for each player
GameTitle	String	Title of the game played
Timestamp	String/Number	When the achievement was recorded
HighScore	Number	Score achieved in the game

Query Need (i): Find all achievements for a specific player sorted by Timestamp

Use: Local Secondary Index (LSI)

- Why LSI?

LSI allows you to query within a single partition key (PlayerID) and sort by a different attribute (Timestamp). Perfect for retrieving a player's achievements in chronological order.

LSI Schema

Attribute	Role in LSI
PlayerID	Partition Key (same as base table)
Timestamp	Sort Key (LSI-specific)

```
Base Table: PlayerStats
+-----------+------------+------------+-----------+
| PlayerID  | GameTitle  | Timestamp  | HighScore |
+-----------+------------+------------+-----------+

LSI: PlayerID-Timestamp Index
+-----------+------------+
| PlayerID  | Timestamp  |
+-----------+------------+
```

Query Need (ii): Create a global leaderboard by GameTitle sorted by HighScore

Use: Global Secondary Index (GSI)

- Why GSI?

GSI allows you to query across all partitions using a different partition key (GameTitle) and sort key (HighScore).

Ideal for building a leaderboard across all players for a specific game.

```
Base Table: PlayerStats
+-----------+------------+------------+-----------+
| PlayerID  | GameTitle  | Timestamp  | HighScore |
+-----------+------------+------------+-----------+

GSI: GameTitle-HighScore Index
+------------+-----------+
| GameTitle  | HighScore |
+------------+-----------+
```

Summary: LSI vs GSI

Feature	Local Secondary Index (LSI)	Global Secondary Index (GSI)
Partition Key	Same as base table	Can be different from base table
Sort Key	Different from base table	Can be different from base table
Query Scope	Within a single partition	Across all partitions
Size Limit	10 GB per partition key	No size limit
Throughput Sharing	Shares with base table	Independent throughput
Use Case Example	Player achievements by time	Game leaderboard by score

4. Cloudwatch

Metrics Collection

- EC2 Instances: CloudWatch collects default metrics like CPU utilization, disk I/O, and network throughput.

- For deeper insights (e.g., memory, disk space), install the CloudWatch Agent.

- S3 Buckets: CloudWatch captures request metrics such as GetRequests, PutRequests, 4xxErrors, and Latency via S3 request metrics.

- Custom Metrics: Application-level metrics (e.g., video encoding time, stream failures) can be pushed using the PutMetricData API.

This enables real-time visibility into infrastructure and application health.

Alarms & Notifications

- Set CloudWatch Alarms on critical metrics (e.g., CPU > 80%, S3 5xx errors).

- Configure SNS (Simple Notification Service) to send alerts via email, SMS, or trigger Lambda functions.

- Example:

- Alarm on EC2 StatusCheckFailed

- Alarm on S3 4xxErrors exceeding threshold

- Alarm on custom metric VideoStreamFailures

- ClouudWatch Dashboards to visualize:

- EC2 performance trends

- S3 access patterns

- Application-level metrics

- Alarm status and thresholds

Log Analysis for Crash Diagnosis

- Use CloudWatch Logs to ingest:

- EC2 system logs

- Application logs (via CloudWatch Agent or Fluent Bit)

5. AWS System Manager

Five AWS Systems Manager Features to Address Operational Challenges

1. Session Manager – Secure Access Without SSH

- Purpose: Enables secure, browser-based or CLI access to EC2 instances and hybrid servers without opening SSH ports or using bastion hosts.

- Benefits:

- No need for SSH keys or exposed ports

- Fully auditable sessions (recorded and stored in S3)

- Ideal for managing instances in private subnets

solves the challenge of secure remote access across global infrastructure.

2. Patch Manager – Consistent Security Patching

- Purpose: Automates the detection, approval, and deployment of OS and software patches across Windows and Linux systems.

- Features:

- Define patch baselines for compliance

- Schedule patching via maintenance windows

- Apply patches to EC2 and on-premises servers

Ensures consistent, policy-driven patching across thousands of nodes.

3.State Manager – Configuration Management

- Purpose: Maintains and enforces desired state configurations for instances (e.g., registry settings, software versions, firewall rules).

- Capabilities:

- Automatically apply configurations on boot or schedule

- Integrates with SSM documents for repeatable tasks

- Tracks compliance status

Helps enforce consistent configurations across environments.

4.Automation – Repetitive Task Execution

- Purpose: Executes runbooks to automate routine tasks like instance provisioning, backup, cleanup, or patching.

- Features:

- Use prebuilt or custom SSM Automation documents

- Trigger workflows manually or via events (e.g., CloudWatch alarms)

- Supports conditional logic and approvals

5. Inventory & Insights – Visibility and Compliance

- Purpose: Collects metadata about instances (e.g., installed software, network config, patch status) and provides compliance insights.

- Benefits:

- Centralized view of fleet health

- Integration with Amazon Q Developer for natural language queries

- Supports hybrid environments

Enables proactive monitoring and governance across EC2 and on-premises servers.

Challenge	Systems Manager Feature	Key Benefit
Secure access without SSH	Session Manager	Encrypted, auditable access
Inconsistent patching	Patch Manager	Automated, policy-driven updates
Configuration drift	State Manager	Enforced desired state
Manual operational tasks	Automation	Scalable, repeatable workflows
Lack of visibility &	Inventory & Insights	Fleet-wide monitoring and

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